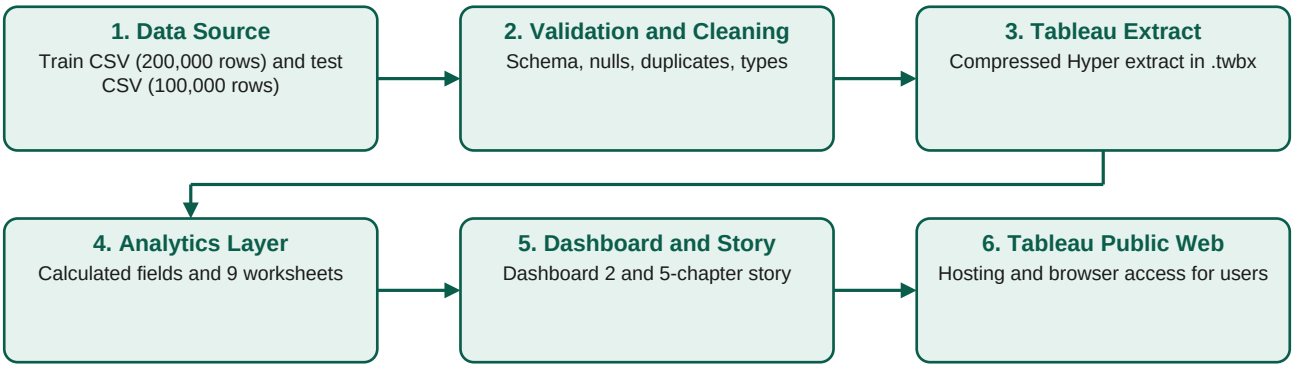
Project Design Phase-I: Solution Architecture

Field	Details
Date	02 November 2026
Team ID	N/A (Individual Project)
Project Name	EV Value & Longevity Analytics Dashboard
Prepared by	Piyush Malviya
Maximum Marks	4 Marks

Solution architecture connects the business problem to the technology that solves it, and shows how data moves from the source files to the end user.

Architecture Diagram



Architecture Layers

Layer	Key Elements	Technology
1. Data Source	37 attributes per vehicle: price, range, battery, safety, sales, rating, country, region	CSV / CSV.GZ files
2. Validation and Cleaning	Null counts, duplicate IDs, totals check, type fixes	Python (pandas), Tableau Data Source tab
3. Storage	Extract for fast filtering, offline development	Tableau Hyper extract (.twbx)
4. Analytics	Value Index, averages, percent of records, filters	Tableau calculated fields
5. Presentation	Nine-chart dashboard, five-chapter story, tooltips	Tableau Desktop (Public edition)
6. Delivery	Web link, interactive filters, PDF or image export	Tableau Public

Component Description

- Data source: the two CSV files hold the same 37 columns, so the same workbook logic is used for both.
- Validation: confirms the total of 1,550,703,961 units and flags five columns with missing values.
- Analytics: replaces raw sums with averages and a Value Index that users can compare directly.
- Delivery: public hosting means no software install, but only public data may be used.